Structural equation modelling

What is SEM?

- SEM is a diverse set of statistical tools that are generally applied in social and behavioural sciences.
- It is used to study complex interrelationships among variables which are not easy to measure, such as intelligence, phobias, anxiety, happiness, perceived ease of use, etc.
- SEM is heavily used in consumer marketing.
- Regression models that you have learnt in the previous lectures also fall under SEM amongst other models such as confirmatory factor analysis (CFA), path analysis etc.

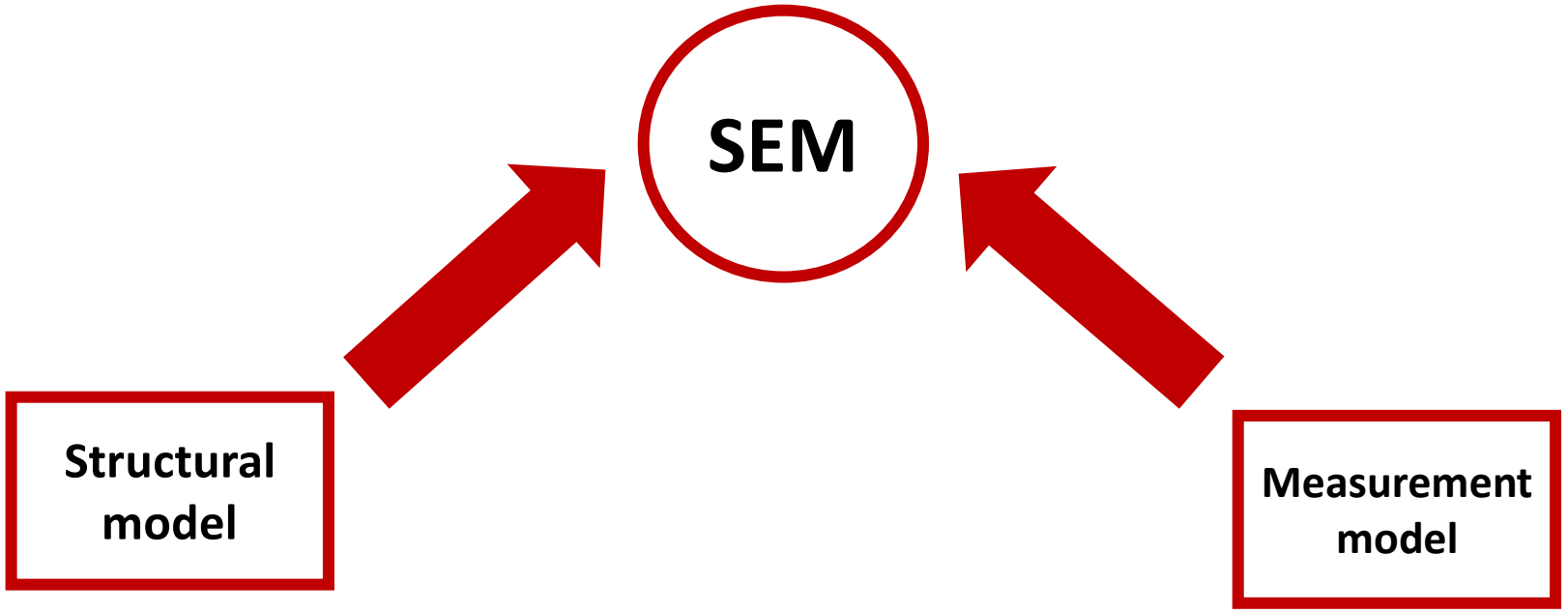


What is SEM?

There are no clear definition of SEM, but it can be decomposed into two sub-models:

(a) a *structural model* (aka **path analysis**)

(b) a *measurement model* (aka **factor analysis**)



What is SEM?



Structural model

Path analysis

- Defines ***complicated relationships*** among variables.
- It specifies the manner by which particular variables directly or indirectly influence (i.e., “cause”) changes in the values of certain other variables in the model.

Measurement model

Factor analysis

- Defines relations between the ***unobserved*** and ***observed*** variables.
- The observed variables are called **measured** variables and the unobserved variables are called **latent variables**.

Structural model: path analysis

- Path analysis is an *extension of multiple regression.*
- It goes beyond regression in that it allows for the analysis of more complicated models.
- Path analysis can be used to analyze models that are more complex (and realistic) than multiple regression.

Structural model: path analysis

Multiple regression:

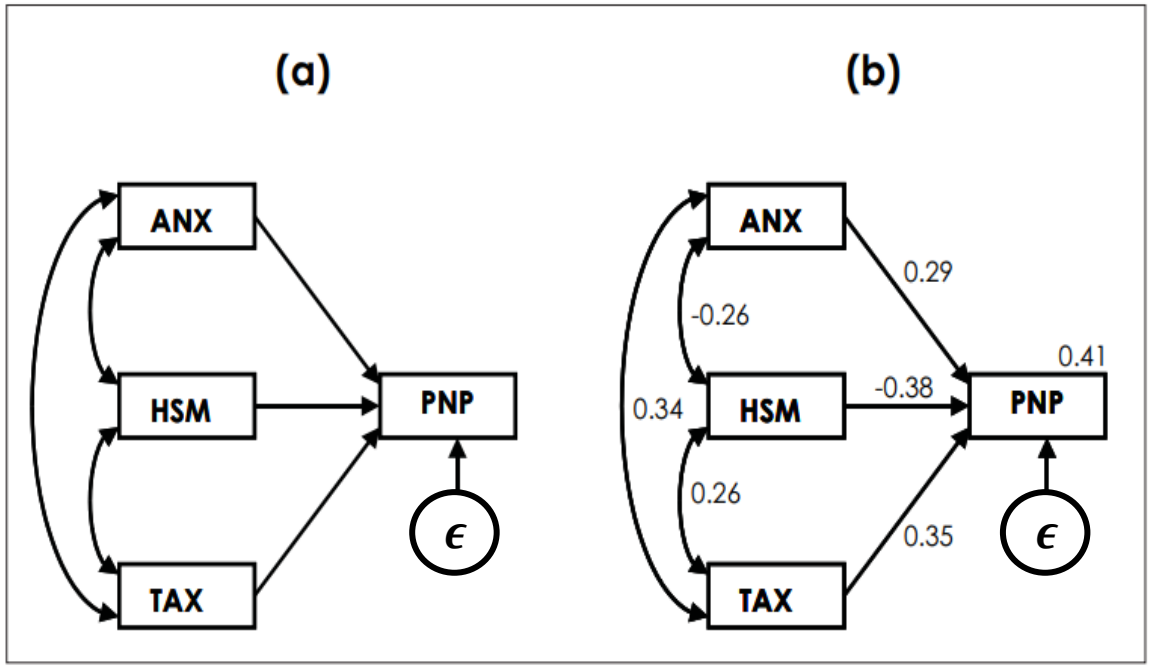
- ***Photonumerophobia:*** is the fear that our fear of numbers will come to light.
- We may want to see whether a person's score on a scale of photonumerophobia (the PNP scale, which ironically is scored using numbers) can be predicted from 3 factors:
  1. overall level of anxiety (ANX),
  2. high school math grades (HSM), and
  3. the discrepancy between what the person estimated last year's taxes to be and what the actual tax was (TAX).

$$\boldsymbol{PNP = \beta_0 + \beta_1 ANX + \beta_2 HSM + \beta_3 TAX + \epsilon}$$

Streiner, D. L. (2005). Finding our way: an introduction to path analysis. *The Canadian Journal of Psychiatry*, *50*(2), 115-122.

Structural model: path analysis

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Structural model: path analysis

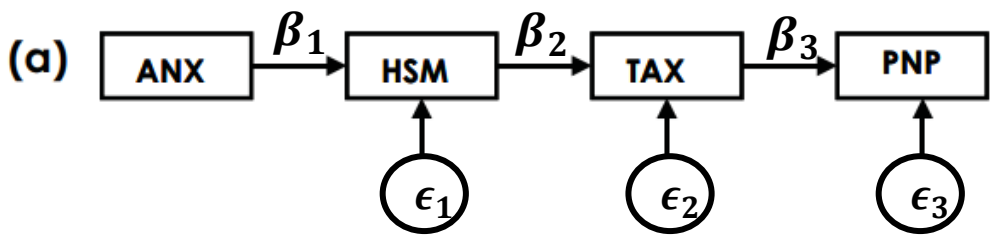
- Using path analysis we can postulate other hypotheses about the relations among the variables and see whether they are better or worse in accounting for the variance in the PNP scale.
- The author found out that the R^2 of the model in multiple regression model was rather low and they hypothesized two new models

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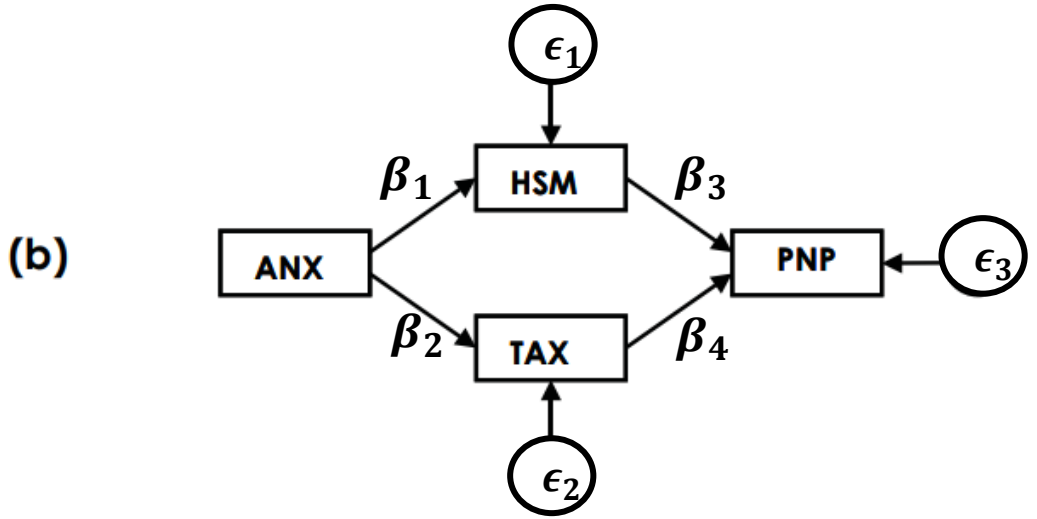
Structural model: path analysis

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In Figure (a), it's hypothesized that, rather than the variables acting separately on PNP, there is truly a path, with anxiety leading to poor math grades in high school, which results in mistakes in completing tax forms and culminates in photonumerophobia.



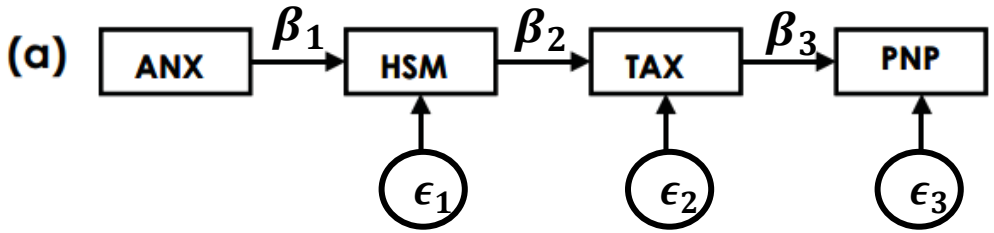
In Figure (b), the hypothesis is that anxiety results in both poor grades and computational errors in filling out taxes and that both of these together lead to photonumerophobia.



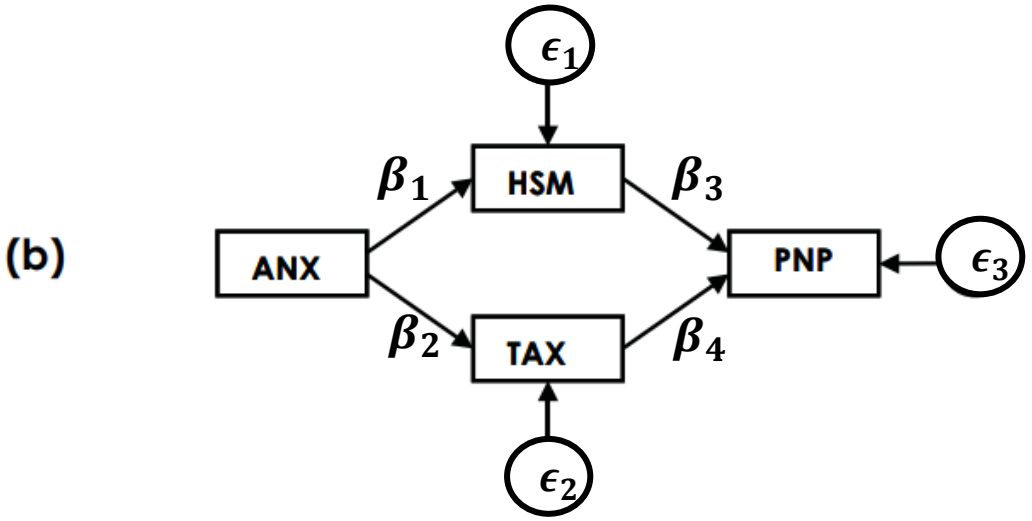
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Structural model: path analysis

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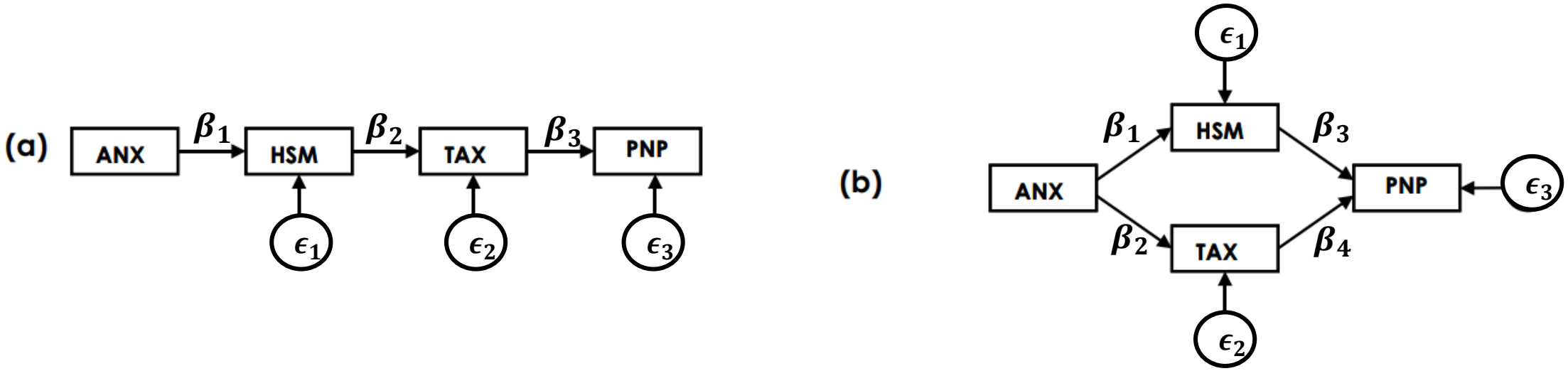
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More formally we say that the effect of ANX on PNP is **mediated** by HSM and TAX through the **structure or path hypothesized**. Hence the name structural model or path analysis.

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Structural model: path analysis



- The β's are *identical to the weights of multiple regression*.
- The β's should *be statistically significant*. **Check the p-values!**
- There are several fit indices that tell you how well the model fits the data, we will discuss later.

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Measurement model: factor analysis

- Measurement model involves variables that cannot be measured directly (**latent variables**) and are generally inferred using other measurable (**observed**) variables.
- Latent variables are applied to concepts such as intelligence, anxiety, depression, quality of life, coping style, schizophrenia, locus of control, and hundreds of others we encounter every day in psychiatry and psychology.

Streiner, D. L. (2006). Building a better model: an introduction to structural equation modelling. *The Canadian Journal of Psychiatry*, *51*(5), 317-324.

Measurement model: factor analysis

- Let us focus on anxiety. When we say that a person is anxious, what we mean is that he or she is showing observable behaviours that are manifestations of one or more of these facets.
- In the *physiological realm*, for instance, there may be tachycardia, shortness of breath, and sweatiness (all of which are measurable).
- In the *cognitive realm*, there may be decreased ability to concentrate and hypersensitivity to perceived threats (again measurable).
- In other words, anxiety is something we hypothesize to tie together observable phenomena that are correlated with each other to some degree.
- Therefore, anxiety is a latent variable which is measure through some observed variable such as heart rate, poor concentration etc. You can think of the latent variables as hidden variables.
- According to one theory*, anxiety has 4 facets—cognitive, affective, behavioural, and physiological—that are themselves unobservable hypothetical constructs.

*Streiner, D. L. (2006). Building a better model: an introduction to structural equation modelling. *The Canadian Journal of Psychiatry*, *51*(5), 317-324.

Measurement model: factor analysis

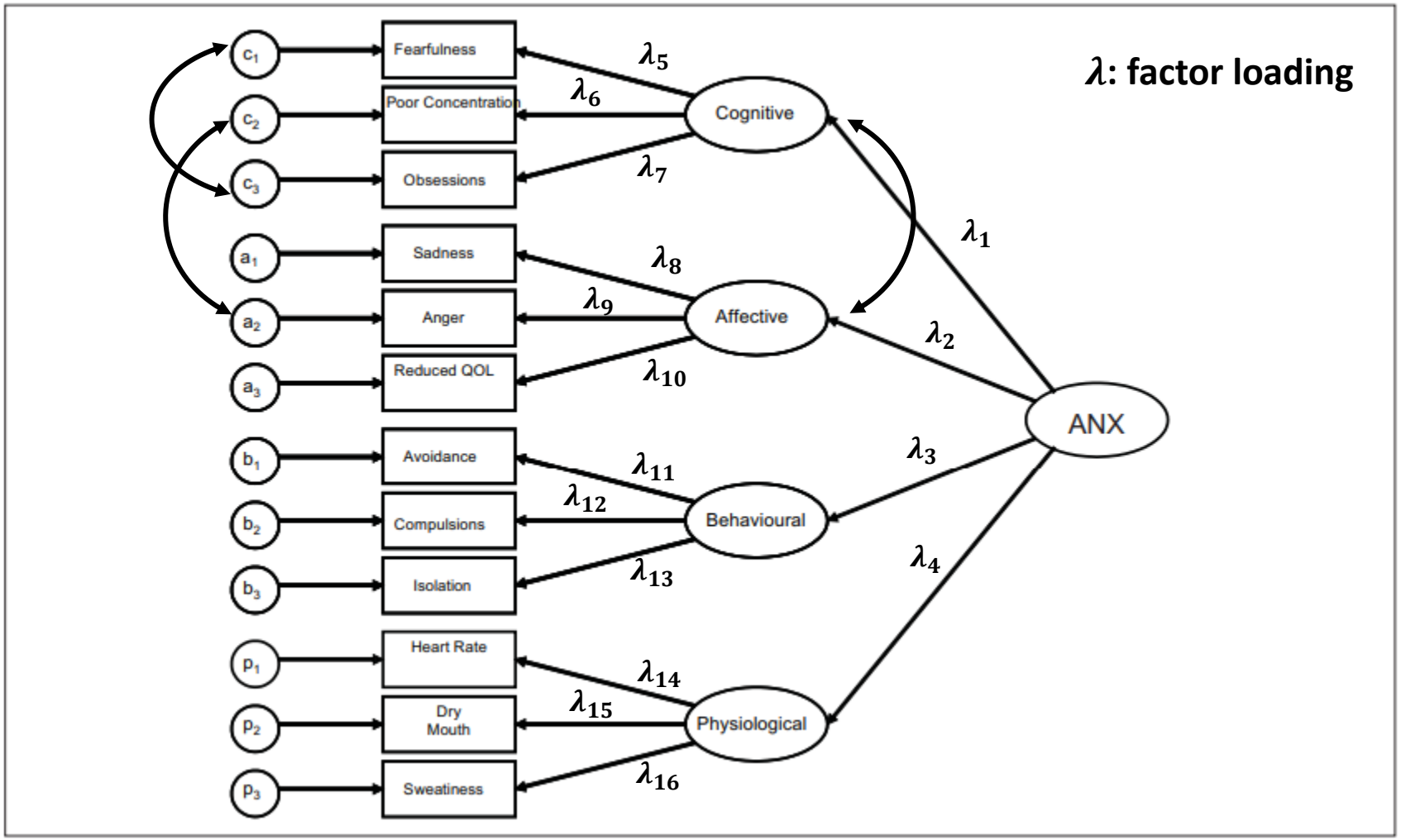
Latent variable

Error

Measured variable

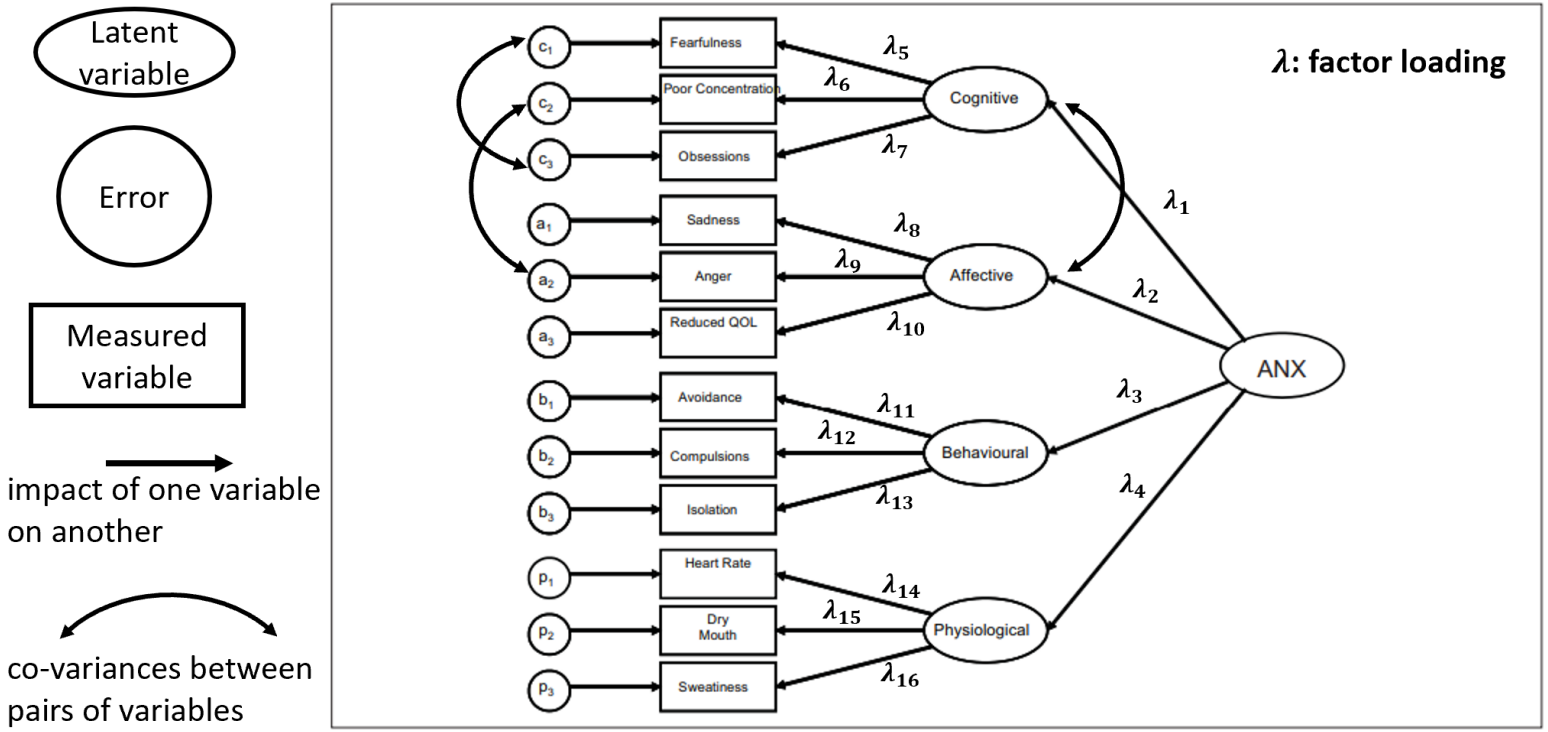
impact of one variable on another

co-variances between pairs of variables



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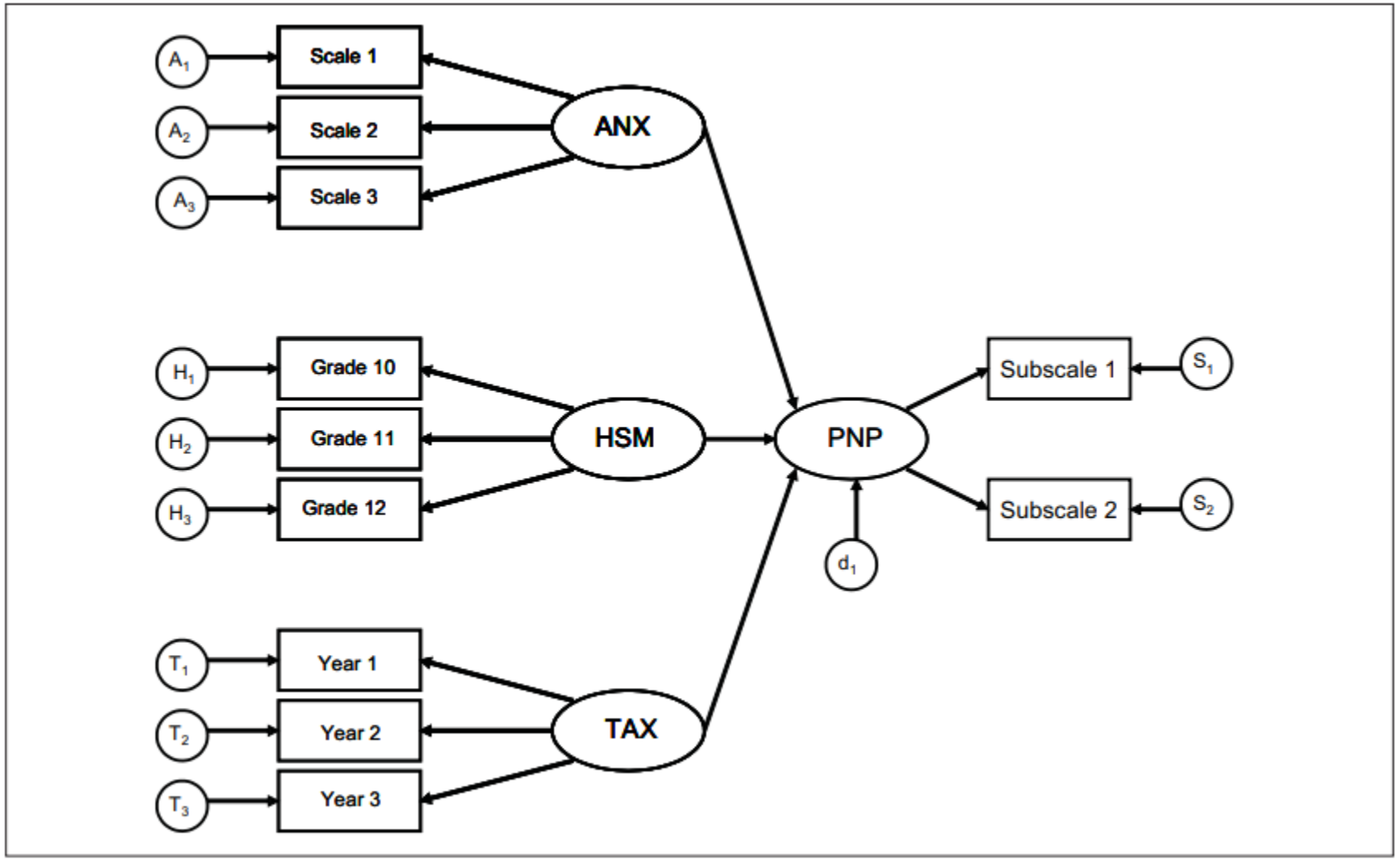
Measurement model: factor analysis



- Generally for the measurement model, researchers have a **theory and conceptualization of the latent variables**. Then they design measurement instruments such as a **questionnaire or test** that provides them with multiple indicators of the latent variable.
- The elements of the measurement instruments form the measured/observed variables.

SEM for Photonumerophobia

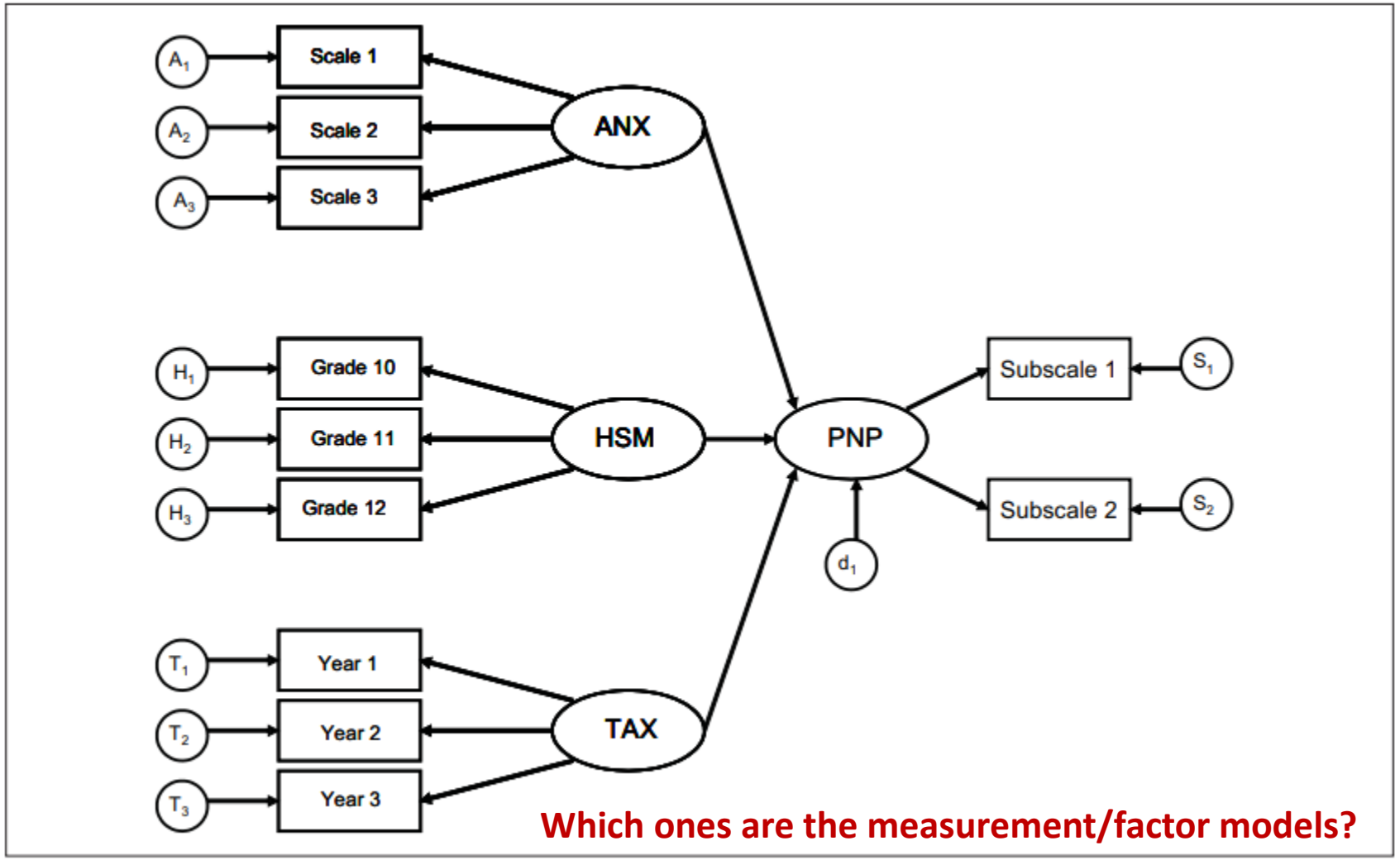
Predict a subject's degree of PNP on the basis of ANX, HSM, and TAX



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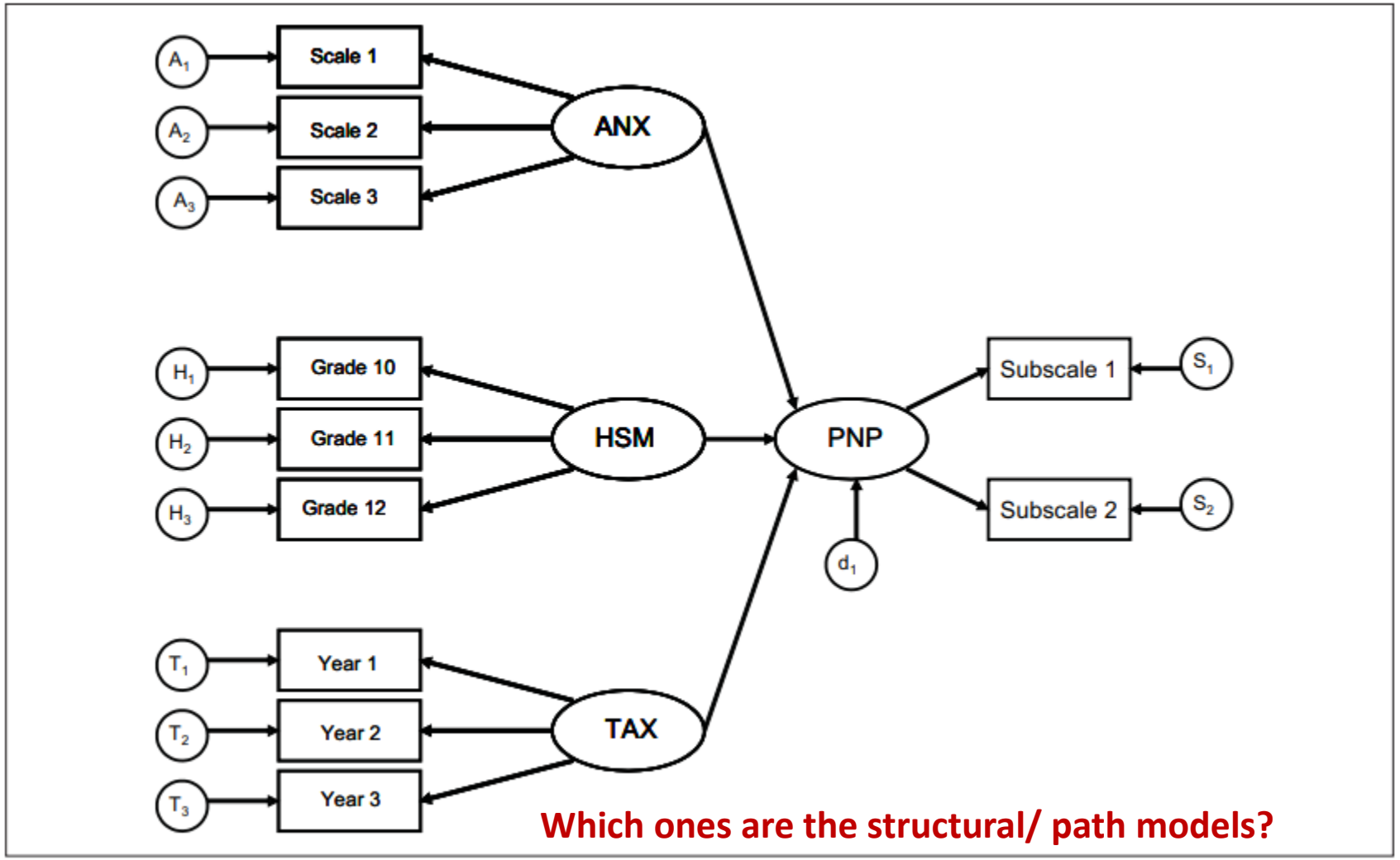
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SEM: model specification

- The first step in SEM is called the ***model specification*** stage. In this step we specify the model variables and it is probably the most difficult—and most important—part.

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- The **primary cause of poorly fitting models is the omission of crucial variables**, and there are no programs in the world that can help us in this regard.
- If the prime determinant of PNP is actually the PNP level of one's parents (because of either genetics or learning), and if this isn't correlated with any of the other variables we're examining, our model will explain little of the variance, and we will never know why.

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SEM: model fitting

- This step is an easy step. Generally we use a computer program to fit the hypothesized SEM model to the data.
- This step involved computing all the model parameters $\boldsymbol{\theta} = [\boldsymbol{\lambda},\ \boldsymbol{\beta}, \boldsymbol{\Sigma}]$: where $\boldsymbol{\lambda}$ is the vector to all the loading factors and $\boldsymbol{\Sigma}$ is the covariance matrix.
- After computing $\boldsymbol{\lambda}$'s and $\boldsymbol{\beta}'$s:
  - one should ***check if they all have the right sign***.
  - check if they are significant? (check p-values)
  - if the answer to either question is no, it may be best to respecify the model by dropping nonsignificant variables and (or) seeing whether there are variables in your data set that should be included.
- Generally for SEM we use the **maximum likelihood estimation (MLE)** method to estimate the model parameters.
- On a high level, the parameter estimation is done by comparing the population co-variance matrices representing the relationships between variables and the estimated co-variance matrices of the best fitting model.

SEM: model fit estimation

- After the SEM model is fitted to data, one needs to **test the goodness of the model fit**, just like in multivariate regression model we check the R^2 value.
- The basic task in SEM modeling, forming the basis for accepting or rejecting models and, more usually, accepting one competing model over another.
- Assessment of fit essentially calculates how similar the predicted data are to matrices containing the relationships in the actual data.
- There are differing approaches to assess model fit. Because different assessment of fit capture different elements of the fit of the model, it is appropriate to report a selection of different fit measures.

SEM: model fit estimation

- **Chi-squared p-values:** The *null hypothesis* is the difference between the observed covariance matrix and the model co-variance matrix is zero, and *the alternative hypothesis* is the difference is not equal to zero.
  Note: For this test if the $p > \alpha$, the pre-defined significance level, then we fail to reject null and conclude that the model fit is good.

- **Root Mean Square Error of Approximation (RMSEA):** difference between the observed correlation and the predicted correlation. Fit index where a value of zero indicates the best fit. While the guideline for determining a "close fit" using RMSEA is highly contested, most researchers concur that an RMSEA of .1 or more indicates poor fit.

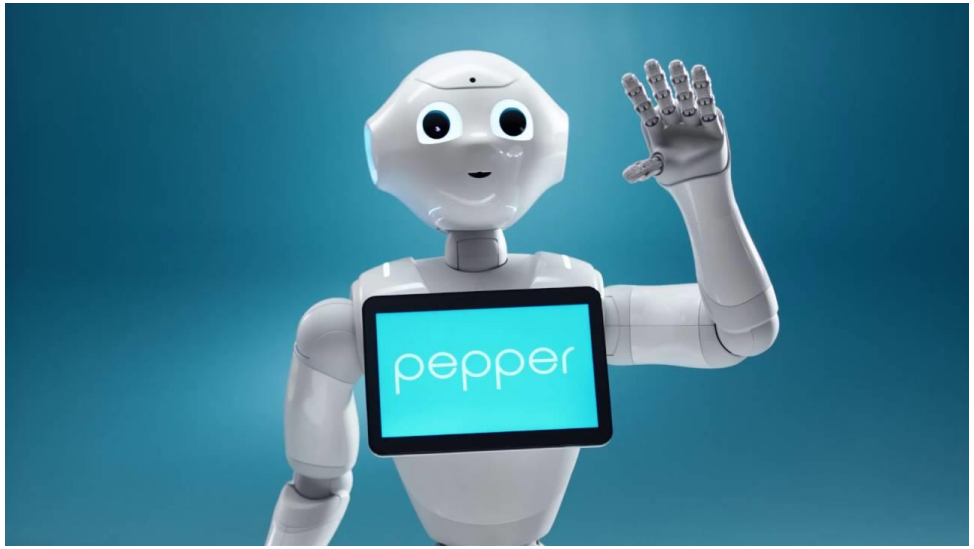
- **Standardized Root Mean Residual (SRMR):** The SRMR is a popular absolute fit indicator and is defined as the standardized difference between the observed correlation and the predicted correlation. A value less than .08 is generally considered a good fit.

- **Comparative Fit Index (CFI):** In examining baseline comparisons, the CFI depends in large part on the average size of the correlations in the data. If the average correlation between variables is not high, then the CFI will not be very high. A CFI value of .95 or higher is desirable.

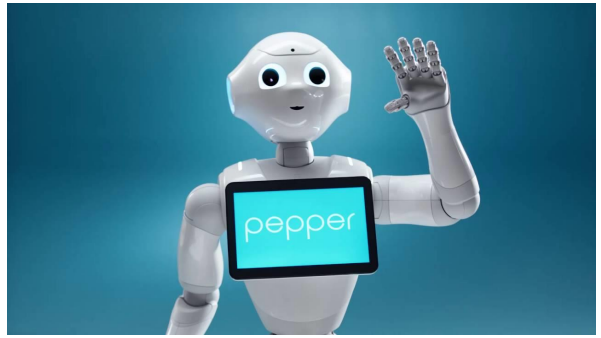
- **Akaike information criterion (AIC):** A test use to compare two competing models**: the preferred model is the one with the lowest AIC value**. $AIC = 2k - 2\ln(\mathcal{L})$, where k is the number of parameters in the statistical model, and $\mathcal{L}$ is the maximized value of the likelihood of the data given the model parameters.

SEM example

What factors influence human robot initiation of interaction?



SEM example



Openingsstrategie
Motivators / Antecedents:

- Gaze (Aankijken)
- Motion (Beweging)
- Language (Taalgebruik)
- Locomotion (Voortbeweging)

Mediators:

- Perceived Ease of Use
- Perceived Usefulness
- Arousal
- Technophobia

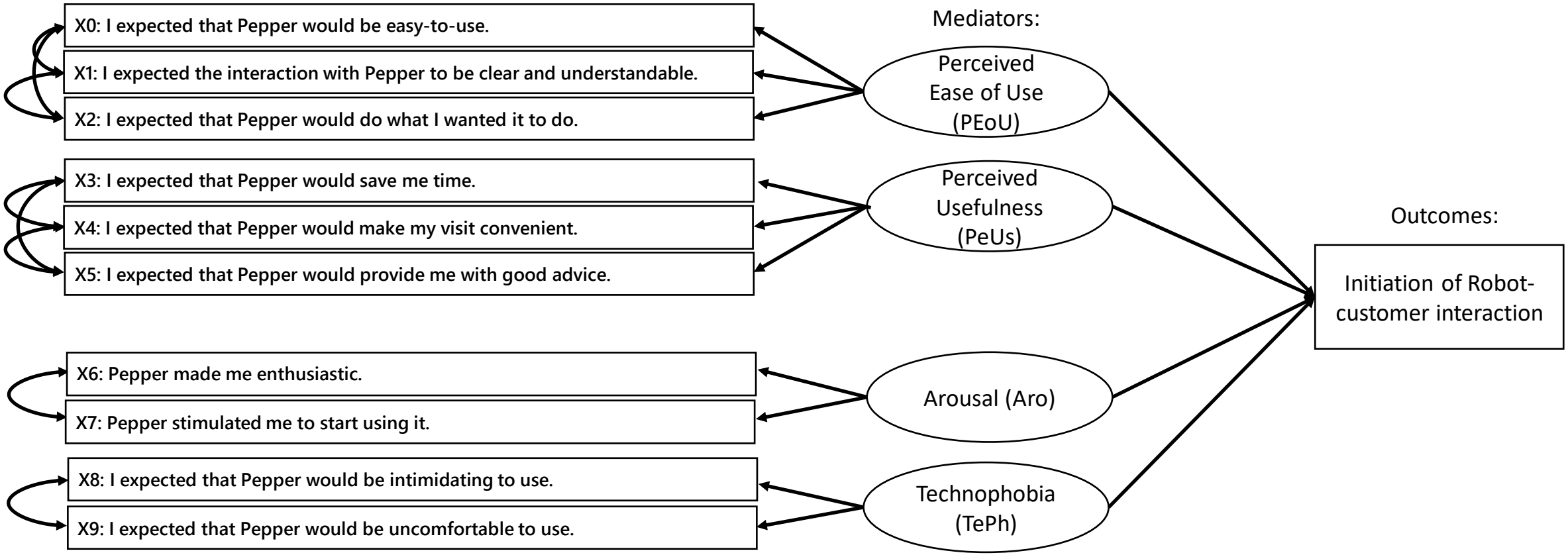
Outcomes:

- Initiation of Robot-customer interaction

Moderators:

- Age
- Gender
- Cultures
- Education

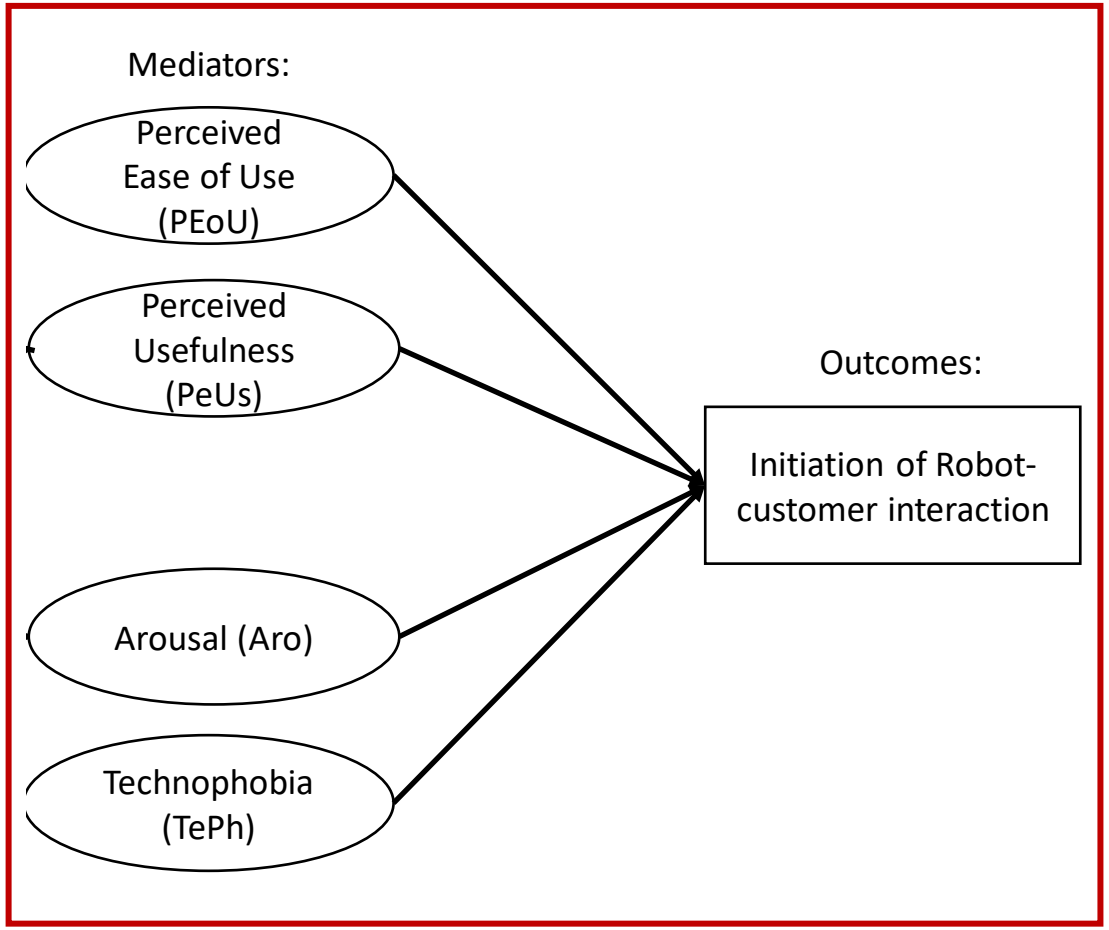
SEM example



Amsterdam University of Applied Sciences

SEM example

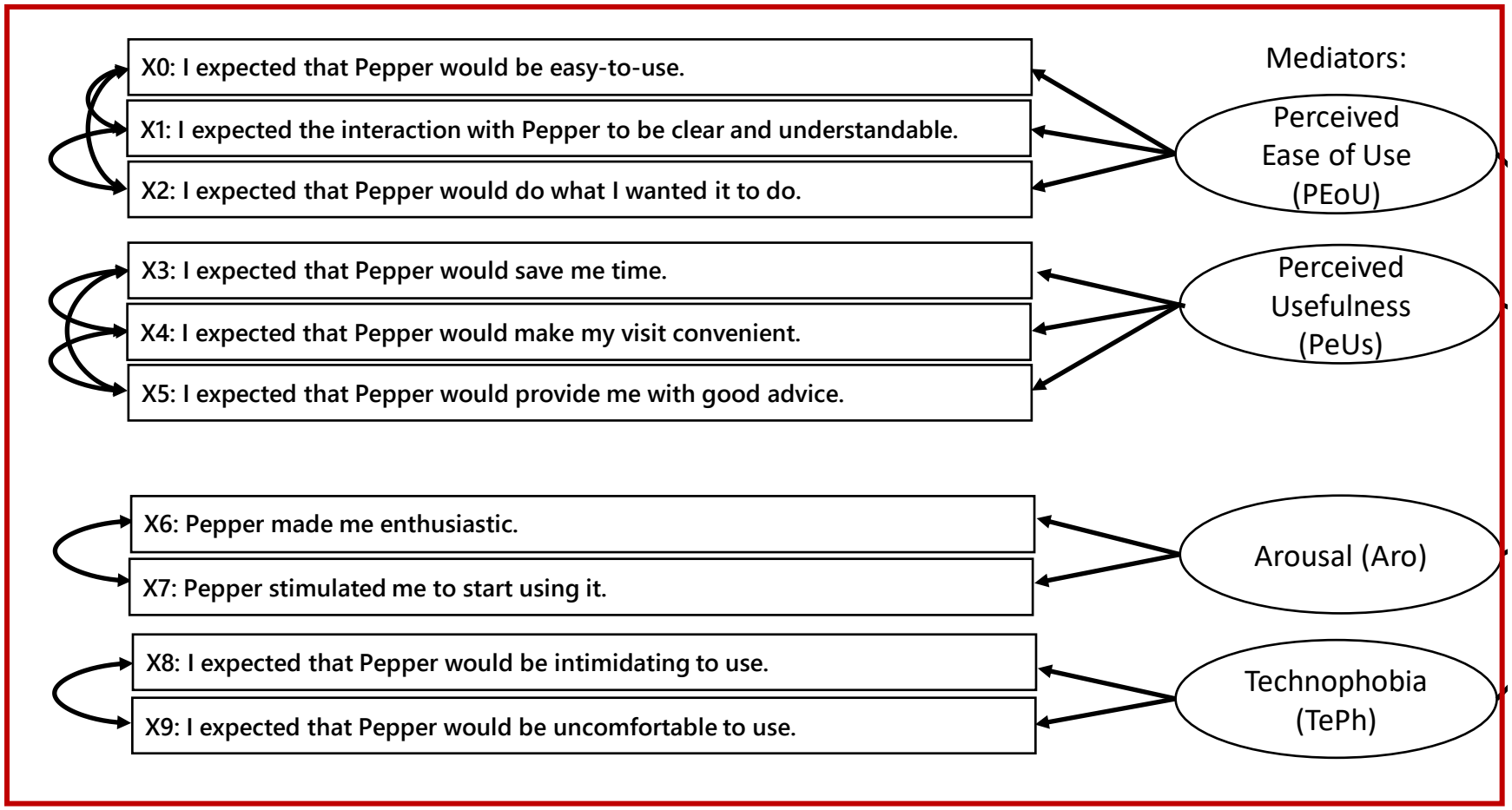
Structural model



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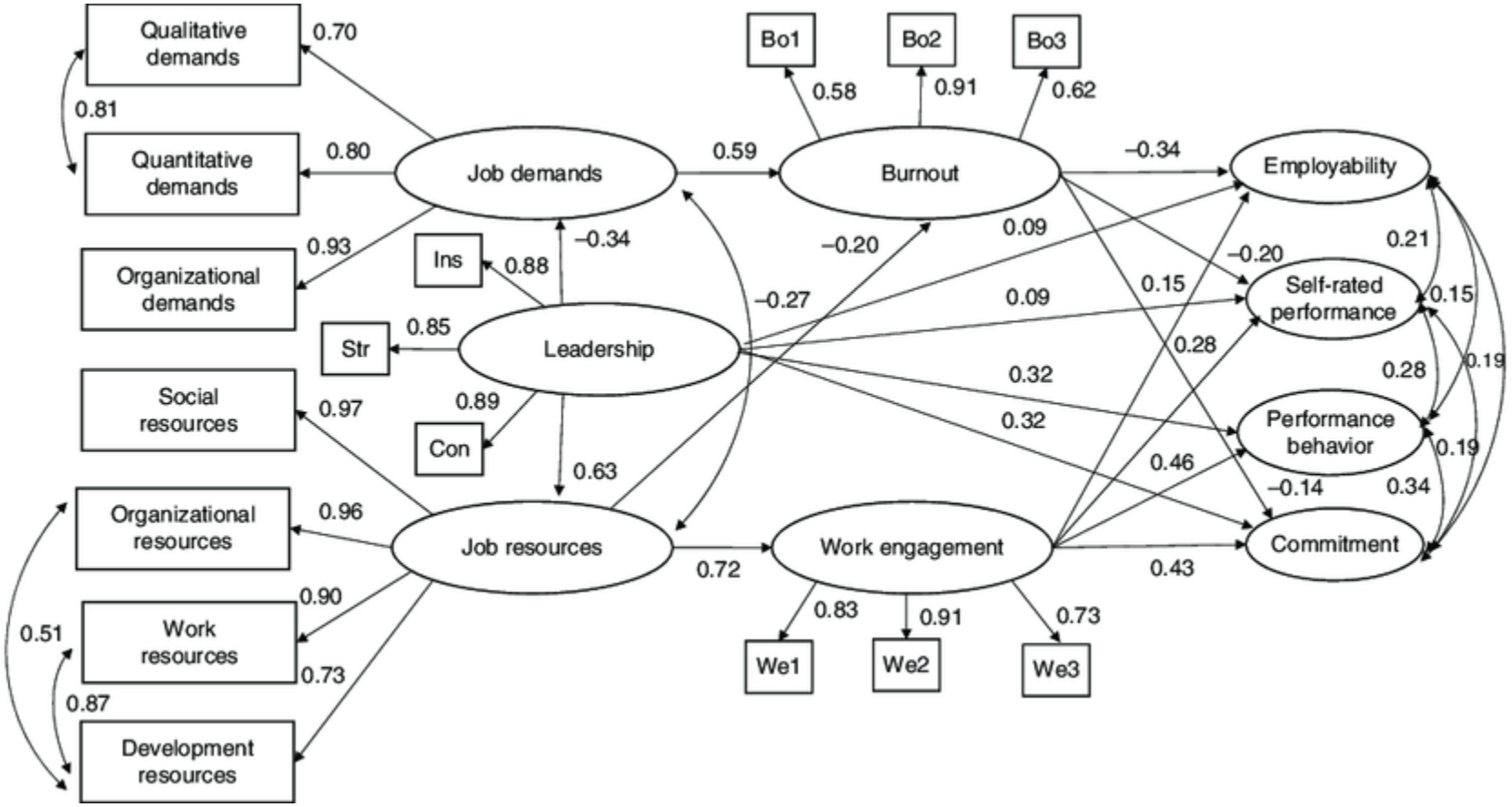
SEM example

Measurement model



SEM example

The authors in this study integrated leadership into the job demands-resources model and argued that engaging leaders who inspire, strengthen, and connect their followers would reduce employee's levels of burnout and increase their levels of work engagement.



They inferred from an online survey of a representative sample of the Dutch workforce (n = 1, 213) that leadership only had an indirect effect on burnout and engagement – via job demands and job resources – but not a direct effect.

Wilmar B Schaufeli. Engaging leadership in the job demands-resources model. Career Development International, 2015.